

Assignment 2

cc371: Artificial Intelligence

SST, UMT

Student's Name: _____

Student Id: _____

Total Marks: 100

Note1: Upload the folder (with your ID as its name) containing your code files and assignment report on LMS by the deadline.

Note 2: For Q1, the name of your matlab code file is 'q1.m', for Q2, the name of your matlab code file is 'q2.m' and so on

Note 3: One of the good viewer is Irfanview. Download from <http://www.irfanview.com/> or Locally

(You can convert images from other formats to the pgm or ppm format by opening them in Irfanview and then saving in the desired format)

Note 4: 5% marks are allocated for posting at least one query on the discussion forum or posting a reply to one query on the discussion forum corresponding to this assignment.

1) (10) Write a matlab code to read the files (owl.pgm and mecca06.pgm) and write these files with the following names.

owl.pgm should be written as owl_Out.pgm

mecca06.pgm should be written as mecca06_Out.pgm

2) (30=5+5+....+5) Write a matlab program that do the following:

- a) Reads the files (owl.pgm and mecca06.pgm) in a matrix m1, m2, m3 etc.
- b) Add 50 to each pixel of the matrix (m1, m2 etc.) using for loop. If pixel value (after adding 50) becomes greater than 255, then assign 255 to that pixel value. Save the matrix in a files (owl_IntensityInc50_Out.pgm, mecca06_IntensityInc50_Out.pgm)
- c) Take the logarithm (log) of base 10 of each pixel of the matrix (m1, m2 etc.) and save the matrices in the files (owl_Log_Out.pgm, mecca06_Log_Out.pgm)
- d) Add 20 to each pixel of the matrix (m1, m2 etc.) using for loop and save it as owl_Intensity20_Out.pgm and mecca_Intensity20_Out.pgm.
Add 40 to each pixel of the matrix (m1, m2 etc.) using for loop and save it as owl_Intensity40_Out.pgm and mecca_Intensity40_Out.pgm.
Add 60 to each pixel of the matrix (m1, m2 etc.) using for loop and save it as owl_Intensity60_Out.pgm and mecca_Intensity60_Out.pgm.
Add 80 to each pixel of the matrix (m1, m2 etc.) using for loop and save it as owl_Intensity80_Out.pgm and mecca_Intensity80_Out.pgm.
Add 100 to each pixel of the matrix (m1, m2 etc.) using for loop and save it as owl_Intensity100_Out.pgm and mecca_Intensity100_Out.pgm.
- e) Repeat part b and part c without using for loop

- 3) (15) Write a matlab code to read the files owl.ppm in matrix mR1, mG1 and mB1. Similarly, read the file mecca06.ppm in matrix mR2, mG2 and mB2. Write these matrices in the files with the following names.

owl.ppm should be written as owl_Out.ppm

mecca06.ppm should be written as mecca06_Out.ppm

- 4) (45= 5+ 5++5) Write a matlab program that do the following:
- Reads the files (owl.ppm and mecca06.ppm) in a matrix mR1, mG1, mB1, mR2, mG2, mB2 (in general m means matrix, R means Red layer, G means green layer, B means Blue layer, 1 means first image i.e., owl.ppm and 2 means second image i.e., mecca06.ppm)
 - Make each pixel of the G matrix (matrix containing green layer) and B matrix of both the images to 0 (i.e., make the matrices m1G, m1B, m2G and m2B to zero). Write these three matrices (of red, green and blue layer) of each image in a file named owl_Red_out.ppm
 - Make each pixel of the R matrix (matrix containing red layer) and B matrix of both the images to 0 (i.e., make the matrices m1R, m1B, m2R and m2B to zero). Write these three matrices (of red, green and blue layer) of each image in a file named owl_Green_out.ppm
 - Make each pixel of the R matrix (matrix containing red layer) and G matrix of both the images to 0 (i.e., make the matrices m1R, m1G, m2R and m2G to zero). Write these three matrices (of red, green and blue layer) of each image in a file named owl_Blue_out.ppm
 - Make each pixel of the R matrix (matrix containing red layer) to zero. Write these three matrices (of red, green and blue layer) of each image in a file named owl_GreenBlue_out.ppm
 - Make each pixel of the G matrix (matrix containing green layer) to zero. Write these three matrices (of red, green and blue layer) of each image in a file named owl_RedBlue_out.ppm
 - Make each pixel of the B matrix (matrix containing blue layer) to zero. Write these three matrices (of red, green and blue layer) of each image in a file named owl_RedGreen_out.ppm
 - Take the logarithm (log) of base 10 of each pixel of the three (red, green and blue) matrices of the images and save the matrices in the files (owl_Log_Out.ppm and mecca06_Log_Out.ppm)
 - Generate the gray scale image of these colored images i.e., owl.ppm and mecca.ppm. Take the average of pixels of these three matrices of each image and generate one matrix. Later save this matrix in a file named owl_grayScale_Out.pgm and mecca06_grayScale_Out.pgm

Report: Your project report should contain your matlab code, the explanation of code/algorithm in simple English, input images and output images.

If you have discussed (only discussed not copied) your assignment with any other student, then write the name of your discussion partner at the top of your report. The template of the assignment report is also uploaded with this assignment.